PlaceIQ

A College Placement Management System

with database-enforced placement rules, analytics and a placement-probability model

Database Management Systems: Digital Assignment 3 (Project Report)

[Course Code] – [Course Name]

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Abstract

Campus placement in most colleges is still coordinated through spreadsheets, forms and e-mail, so the rules that matter most (who is eligible for a drive, whether a student may hold two accepted offers, which status changes are legal) live in people's heads or in scattered application code. This project, PlaceIQ, is a placement management system whose central design decision is to enforce those rules **inside the database**. A PostgreSQL 16 schema of 13 tables in third normal form carries six triggers, two stored functions, one stored procedure, six views (three of them built on window functions) and a materialized view. Invalid writes are refused by the database itself with custom error codes (PL001–PL008), whether they come from the web application or from a raw SQL session; one of them keeps every application on or before its drive's deadline. A Flask backend written with plain SQL (psycopg2, no ORM) exposes 36 REST endpoints, runs offer acceptance as an explicit transaction, and serves a React front end with a student portal, an admin console and an analytics dashboard. A logistic-regression model estimates each student's placement probability. The schema was also ported to Oracle 21c and run end to end in SQL*Plus. The whole system is served from one address (http://localhost:8080): with one Docker command, as a Windows desktop app, or on a temporary public link.

On a synthetic dataset of 180 students, 25 drives and 517 applications, an end-to-end smoke test passed 48 of 48 checks, an automated check of the API documentation passed 161 of 161, and browser checks through the single address passed 12 of 12 in each way of running it. All 10 date-ordering rules hold with zero violations. Every business rule was shown to be refused at the database level, including two simultaneous offer acceptances, which the trigger's row lock serialized. On a 250,000-application benchmark database the indexes cut query time by between 1.8× and 81×. The placement model reached a test accuracy of 0.750 and a ROC-AUC of 0.774, but on a test set of only 36 synthetic students, so it demonstrates the pipeline rather than real predictive power.

Keywords: relational database, triggers, stored procedures, transactions, window functions, materialized views, indexing, placement management, logistic regression.

Table of Contents

Abstract 1
1 Introduction 3
1.1 Literature Review 3
1.2 Problem Statement 4
1.3 Objectives 4
1.4 Scope 4
2 Modules (Implementation and Experimental Setup) 6
2.1 System Architecture 6
2.2 Database Design 6
2.3 Database Logic: Triggers, Routines and Views 8
2.4 Backend API Module 9
2.5 User Interface Module 10
2.6 Placement-Probability Module 11
2.7 Oracle Port 11
2.8 Experimental Setup 12
2.9 Deployment 12
3 Results and Discussion 14
3.1 Functional Verification 14
3.2 Rules Enforced by the Database 14
3.3 Transactions and Concurrency 15
3.4 Placement Analytics 15
3.5 Effect of the Indexes 18
3.6 Placement-Probability Model 18
4 Conclusion and Future Work 20
References 22
Appendix A Code Listings 24
Appendix B Screenshots 28
Appendix C Evidence Files 33

1 Introduction

Every placement season a college's training and placement office (TPO) runs dozens of recruitment drives. Each drive has its own eligibility rules (a minimum CGPA, a maximum number of active backlogs, the branches it is open to), and each applicant moves through a pipeline of shortlisting, interview rounds and offers. The office has to answer two kinds of question at once: operational ones ("who is eligible for this drive?", "which applicants are still pending?") and analytical ones ("what share of each department has been placed?", "which recruiters made the most offers?").

PlaceIQ is a web-based system for this workflow, built for the DA-3 project of the Database Management Systems course. Its emphasis is the database: the schema, the rules it enforces, the queries behind the analytics, and the transactions that keep related tables consistent.

1.1 Literature Review

The relational model and normalization. Codd's relational model [1] proposed representing data as relations and introduced a normal form for them, separating how data is logically organized from how it is stored. Normalization to third normal form, as presented in standard texts such as Silberschatz, Korth and Sudarshan [3], removes partial and transitive dependencies so that each fact is stored once. PlaceIQ's schema follows this: department and company details are kept in their own tables instead of being copied onto every student and drive row.

Transactions. Härder and Reuter [2] set out the terminology of transaction-oriented recovery and the properties now known as ACID: atomicity, consistency, isolation and durability. Accepting an offer in PlaceIQ updates three tables, and atomicity is what guarantees they never disagree. PostgreSQL's documentation describes how its transactions [10] and row-level locks [11] provide these guarantees in practice, and how trigger functions [8] can enforce rules on every write.

Existing placement management systems. Santhosh Kumar and Srividhya [7] describe a web application that automates registration, record updates, searching and notifications for an engineering college's training and placement unit. Systems of this kind digitize the office's records, but the rules are typically checked in application code, which a direct database edit or a second client can bypass.

Predicting placement. Several studies predict whether a student will be placed from academic records. Shukla et al. [5] built a logistic-regression model that estimates a student's placement probability from historical data with known outcomes. Kumar et al. [6] compared several machine-learning methods on students' academic performance and found Random Forest the most effective. PlaceIQ uses logistic regression, implemented with scikit-learn [4], because its coefficients can be read directly and explained to a student. The prediction is a supporting feature here, not the focus.

Gap addressed. The reviewed placement systems treat the database mainly as storage. PlaceIQ moves the placement rules into the database (triggers and stored routines), adds analytics

computed with window functions and a materialized view, and measures the effect of its indexes instead of assuming it.

1.2 Problem Statement

Design and implement a placement management system in which the placement office's rules are guaranteed by the database, so that no client (the web application, an admin's SQL session or a future integration) can create an inconsistent state. Examples of such states are an ineligible application, two accepted offers for one student, an offer for a rejected applicant, or an application moving backwards in the pipeline. The system must also give students and the placement office the operational and analytical views they need, and its design choices (normalization, indexing, transactions) must be justified with measured evidence.

1.3 Objectives

- Design a normalized (3NF) relational schema for students, departments, skills, companies, drives, applications, interview rounds and offers, with an audit trail of status changes.
- Enforce the placement rules in the database with triggers, a stored function for eligibility and a stored procedure for closing a drive, each failing with a distinct error code.
- Implement offer acceptance and decline as explicit transactions, and show that they are atomic and safe under concurrent requests.
- Provide analytics through views, window functions and a materialized view, and present them in a web dashboard.
- Measure the effect of the indexes with EXPLAIN ANALYZE on a realistically sized dataset.
- Integrate a small, explainable placement-probability model and report its accuracy honestly.
- Port the database layer to Oracle and run it end to end in SQL*Plus.
- Deliver the system from a single address, startable with one command, as a desktop app or on a shareable link, using only free tools.

1.4 Scope

In scope:

- Two roles: a student (register, view eligible drives, apply, track applications, accept or decline offers) and the placement officer (manage drives, move applicants through the pipeline, record interview rounds and offers, close drives, view analytics).
- The PostgreSQL schema, the Flask REST API, the React user interface, the ML service, and the Oracle port of the schema.

Out of scope:

- Real student data and any connection to an institution's academic systems. All data is synthetic, and CGPA is entered by the student at sign-up.

- E-mail or SMS notifications, document uploads (resumes, offer letters) and recruiter logins.
- Running the backend against Oracle: the Oracle port covers the database layer only.

2 Modules (Implementation and Experimental Setup)

2.1 System Architecture

PlaceIQ is a three-tier web application with a separate prediction service (Table 1). The browser talks only to the backend. The backend talks to PostgreSQL with plain SQL through a connection pool and calls the ML service over HTTP when a student asks for their placement probability.

Table 1: Components and technologies

Tier	Technology	Responsibility
Presentation	React 19, React Router 7, Vite 8	Student portal, placement-office console, analytics dashboard (charts drawn as SVG)
Application	Python, Flask 3.0, psycopg2 (no ORM), PyJWT, bcrypt	36 REST endpoints, authentication, role checks, offer transactions
Data	PostgreSQL 16 (Docker)	13 tables, 6 triggers, 2 functions, 1 procedure, 6 views, 1 materialized view
Prediction	Flask, scikit-learn, pandas	Trains the model from the database and serves POST /predict
Delivery	waitress, Docker Compose, Electron, Cloudflare quick tunnel	One port for UI and API; one-command start; desktop app; public link
Oracle port	Oracle Database 21c XE, SQL*Plus	The same schema and rules in PL/SQL

The backend deliberately uses raw SQL rather than an object-relational mapper. Every query the system runs is visible in the source, which matters for a database course. It also means the transaction boundaries are explicit `commit()` and `rollback()` calls rather than behaviour hidden inside a framework.

2.2 Database Design

Figure 1 shows the entity–relationship diagram of the 13 tables, rendered from the Mermaid source in `docs/er-diagram.md`. The central entity is `application`, which resolves the many-to-many relationship between students and drives and carries its own attributes (status and application time). Two further junction tables resolve students↔skills and drives↔eligible departments. A drive with no rows in `drive_eligible_dept` is open to every department.

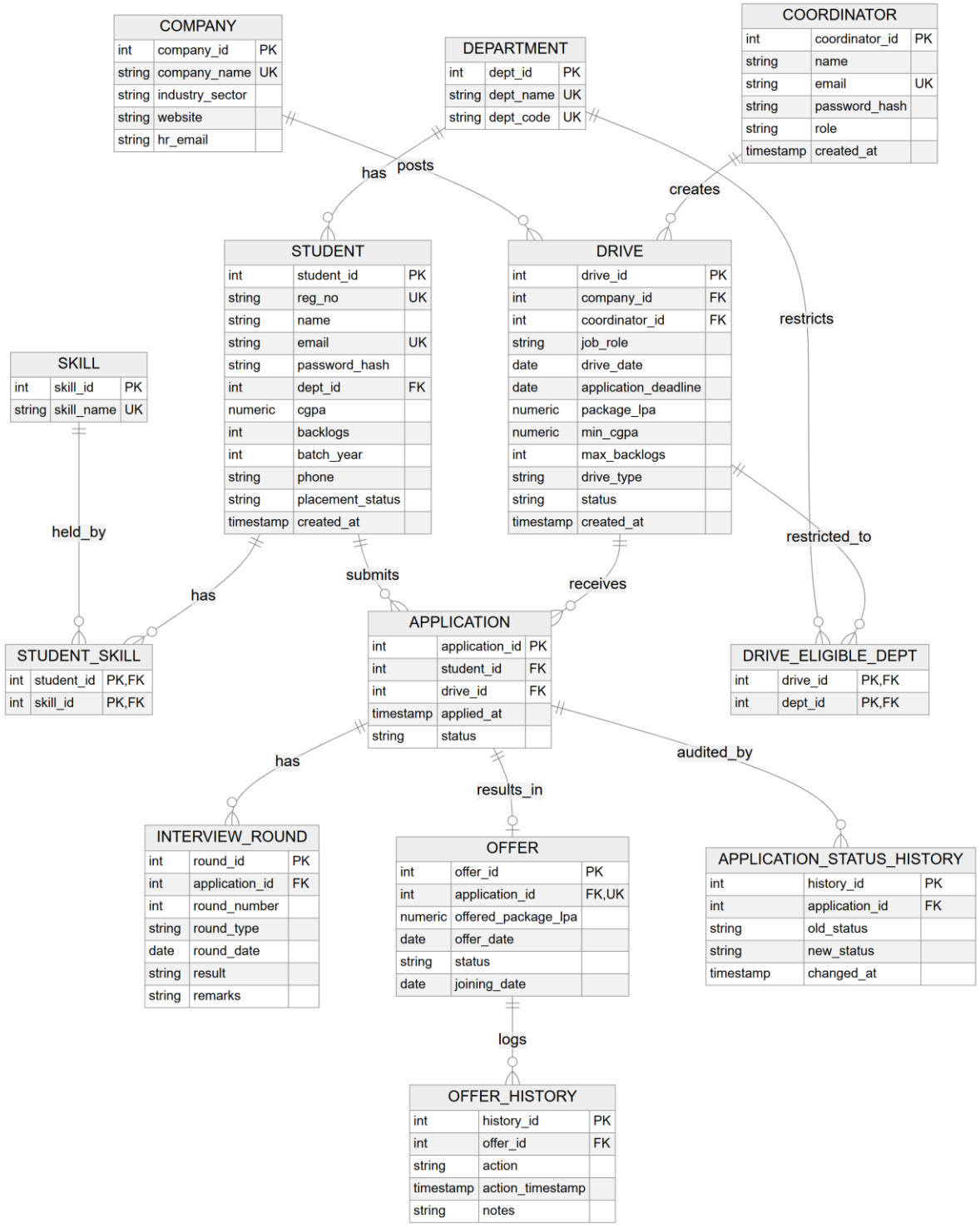


Figure 1: Entity–relationship diagram (13 tables)

Table 2: Tables

Table	Purpose	Key constraints
department	The six branches	UNIQUE name and code
coordinator	Placement-office users (TPO / coordinator)	UNIQUE e-mail; CHECK role
student	Profile, CGPA, backlogs, placement status	UNIQUE reg. no. and e-mail; CHECK $0 \le \text{CGPA} \le 10$; FK department

Table	Purpose	Key constraints
skill, student_skill	Skills and who holds them (M:N)	Composite PK (student, skill)
company	Recruiters	UNIQUE name
drive	One recruitment posting with its cutoffs and application deadline	FK company, coordinator; CHECK status, type, deadline ≤ drive date
drive_eligible_dept	Branches a drive is open to (M:N)	Composite PK (drive, department)
application	A student's application to a drive	UNIQUE (student, drive); CHECK status
interview_round	Rounds of one application	UNIQUE (application, round no.)
offer	At most one offer per application	UNIQUE application; CHECK status
offer_history	Audit log of offer events	CHECK action
application_status_history	Audit log of status changes (written by a trigger)	FK application

All 13 tables are in third normal form: every non-key attribute depends on the whole key and on nothing else. The one deliberate exception is `student.placement_status`, which could be derived from the offers. It is stored because it is read on almost every page, and a trigger keeps it in step with the offer table, so it cannot drift. The full functional-dependency analysis is in `docs/normalization.md`, and every column and constraint, read from the system catalog, is in `docs/evidence/schema-summary.md`.

2.3 Database Logic: Triggers, Routines and Views

The placement rules are implemented as BEFORE triggers that inspect the row being written and raise an error with a custom SQLSTATE code if it breaks a rule (Table 3). Because the check runs inside the database, it applies to every client. The backend translates each code into an HTTP status (`backend/utils/db_errors.py`), so the user sees a meaningful message rather than a generic failure.

Table 3: Database-enforced rules

Code	Trigger / routine	Rule	HTTP
PL001	trg_application_before_insert	A student may apply only if eligible (CGPA, backlogs, branch)	403
PL002	trg_application_before_insert	Applications are accepted only while a drive is Upcoming or Ongoing	409
PL003	trg_offer_before_accept	A student may hold at most one Accepted offer	409
PL004	sp_close_drive	The drive to close must exist	404
PL005	sp_close_drive	A drive can be closed only once	409
PL006	trg_offer_requires_selected	An offer can be recorded only for a Selected application	409
PL007	trg_application_status_transition	Applications move only forward through the pipeline	409
PL008	trg_application_before_insert	An application is dated on or before the drive's deadline, and not in the future	409

The legal status transitions form a small state machine: Applied → Shortlisted / Interview_Scheduled / Rejected; Shortlisted → Interview_Scheduled / Selected / Rejected; Interview_Scheduled → Selected / Rejected; Selected → Offer_Made / Rejected. Offer_Made and Rejected are final. A CHECK constraint cannot express this rule because it sees only the new value. The trigger compares `OLD.status` with `NEW.status`.

Dates. Each drive has an `application_deadline`. "Deadline on or before the drive date" involves one row, so it is a CHECK constraint. "An application is dated on or before its drive's deadline, and not in the future" involves two tables and the clock, so it is part of the application guard trigger (PL008). The database's time zone is set to Asia/Kolkata in `schema.sql`, so a deadline means the end of that day on the college's calendar. In UTC, applications would still be accepted until 05:30 the next morning. A smoke-test failure exposed this, and the setting fixes it.

Two AFTER triggers maintain derived data. `trg_offer_after_insert` marks a student Placed when an offer is recorded. `trg_application_status_audit` writes one `application_status_history` row for every status change, so no code path can change a status without leaving a record.

Rule PL003 and concurrency. If two requests accept two different offers of the same student at the same moment, neither can see the other's uncommitted change, and a naive count of Accepted offers would let both through. The trigger therefore first locks the student's row with `SELECT ... FOR UPDATE`, so the second request waits until the first commits and then sees its result (Section 3.3).

Stored routines. `fn_check_eligibility(student, drive)` returns whether a student meets a drive's rules. It is the single definition of eligibility, used by the application trigger and by `fn_get_eligible_drives(student)`, which returns every open drive a student qualifies for. The procedure `sp_close_drive(drive, INOUT rejected_count)` rejects every still-pending applicant and marks the drive Completed as one unit of work.

Views. Three views serve the dashboards (`placed_students_view`, `company_wise_stats_view`, `department_stats_view`). Three more use window functions: `dept_offer_ranking_view` ranks offers within each department with `RANK()`; `placement_timeline_view` computes a running total with `SUM(...) OVER` and the month-on-month change with `LAG()`; `student_cgpa_standing_view` uses `DENSE_RANK()` and `PERCENT_RANK()` for each student's position within their department [9]. The materialized view `mv_company_dept_placements` stores offers per company and department. It is refreshed on demand with `REFRESH MATERIALIZED VIEW CONCURRENTLY`, which needs a unique index and lets readers continue during the refresh [12], and the dashboard shows when it was last refreshed.

Indexes. Besides the indexes created by primary-key and unique constraints, five indexes support the frequent lookups: `student(dept_id)`, `student(cgpa)`, `application(status)`, `application(drive_id)` and `drive(company_id)` [13]. Their effect is measured in Section 3.5.

2.4 Backend API Module

The Flask backend is organized in blueprints (authentication, students, drives, applications, offers, analytics) and exposes 36 endpoints, each documented in `docs/API.md`. Users authenticate with e-mail and a bcrypt-hashed password and receive a JSON Web Token [14] that carries their role. Every endpoint checks the role, and students can read and change only

their own records. Database access goes through a psycopg2 connection pool [15], with parameterized queries throughout.

The offer transaction. Accepting an offer must update three things together: the offer becomes Accepted with a joining date, the student is marked Placed, and an Accepted row is appended to `offer_history`. `offer_service.accept_offer` runs the three statements on one connection inside `try`, calls `commit()` only after all three succeed, and calls `rollback()` if any statement fails, including when a trigger refuses it. Declining works the same way, and it also sets the student back to Not Placed if no other live offer remains. The full function is listed in Appendix A.

2.5 User Interface Module

The React front end has three areas. Students see an overview (profile, CGPA standing in their department, placement probability), their eligible drives, and their applications. The application detail shows interview rounds, the offer and the status history written by the audit trigger (Figure 2). The placement office manages drives and applicants (Figure 3) and sees an analytics dashboard (Section 3.4). Every chart also has a table view, so no value is available only by hovering.

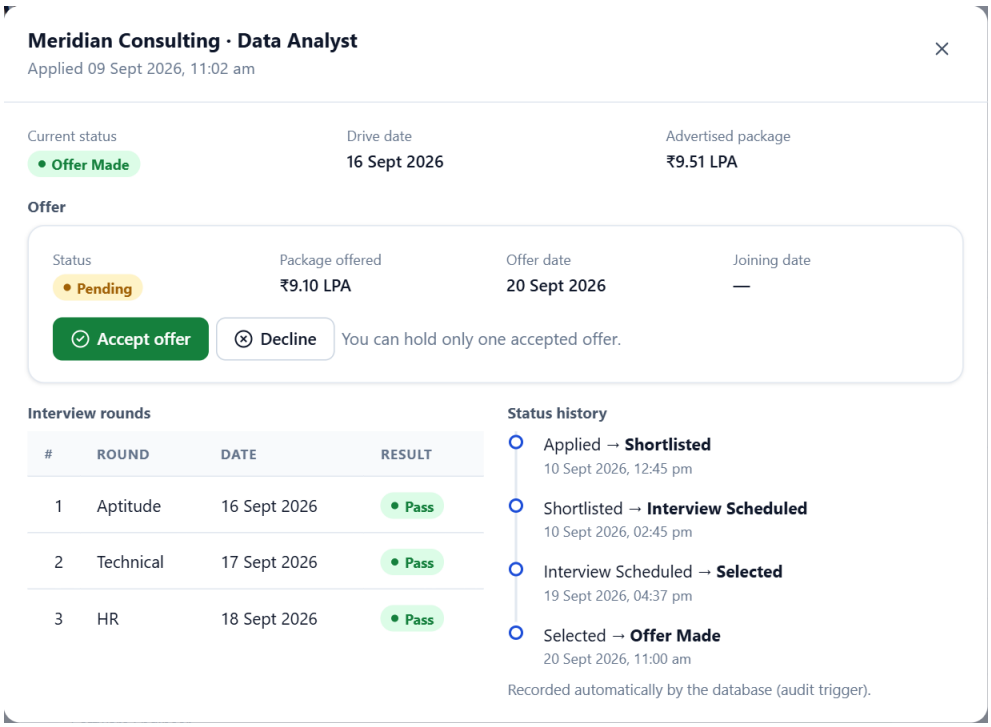


Figure 2: Application detail: offer, interview rounds and the status history recorded by the audit trigger

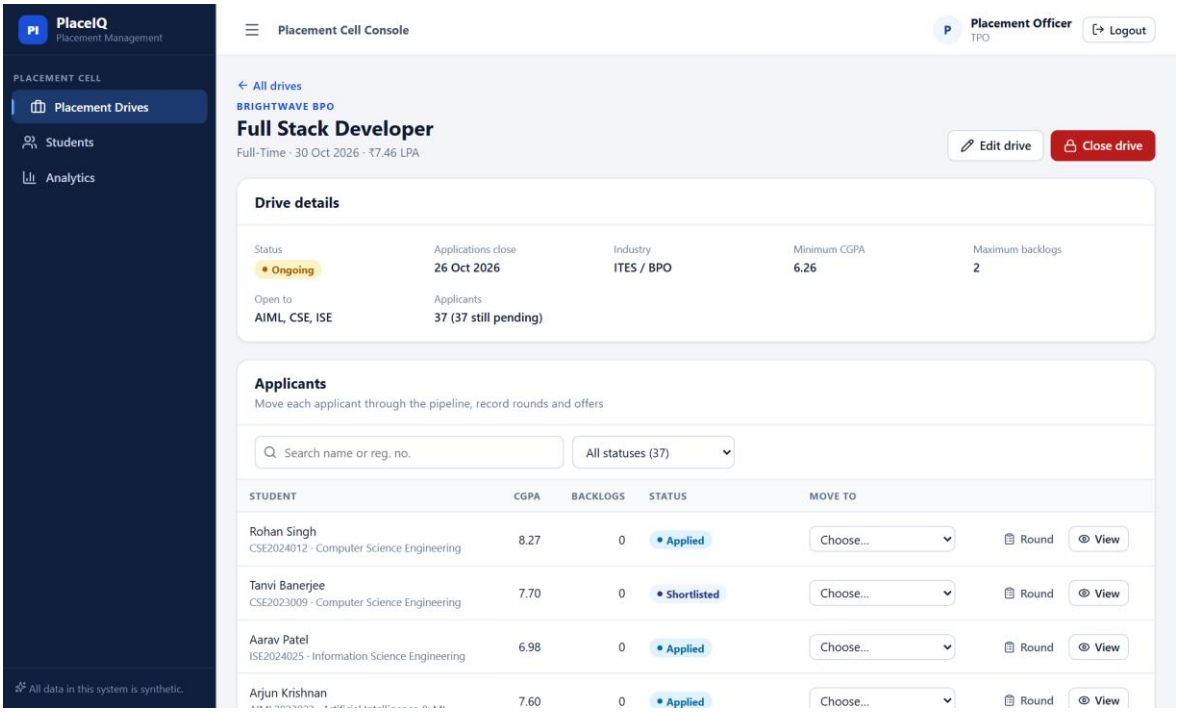


Figure 3: Placement office: a drive's details and its applicants

2.6 Placement-Probability Module

A separate service trains a scikit-learn [4] pipeline: standardized numeric features, a one-hot encoded department, and a logistic-regression classifier. The features are CGPA, number of active backlogs, number of skills (counted with a join on `student_skill`) and department. The label is `placement_status`, which the database trigger sets. The data is split 80/20, stratified by label, with a fixed random seed. The trained model is saved to disk and served by `POST /predict`. The backend calls it for the logged-in student and returns HTTP 503 if the service is unavailable.

2.7 Oracle Port

The schema was ported to Oracle Database 21c Express Edition (`db/schema_oracle.sql`) and run end to end in a single SQL*Plus session (`db/oracle_full_run.sql`). The main differences are:

- Identity columns are implemented with sequences and BEFORE INSERT triggers.
- Rule violations are raised with `RAISE_APPLICATION_ERROR` (−20001 to −20008) instead of custom SQLSTATEs.
- The materialized view is built `REFRESH COMPLETE ON DEMAND`.
- Identifiers are shortened to Oracle's 30-character limit.
- Rule PL003 is a COMPOUND trigger [16]. A row-level trigger on `offer` cannot query `offer` itself (ORA-04091, "table is mutating"), so the compound trigger collects the affected rows and checks them after the statement.

The differences are documented line by line in `docs/oracle-differences.md`.

2.8 Experimental Setup

Table 4: Environment

Item	Value
Machine	12th Gen Intel Core i7-12700H, 15.6 GB RAM, Windows 11
Database	PostgreSQL 16.15 in Docker (postgres:16 image)
Oracle	Oracle Database 21c XE (gvenzl/oracle-xe:21-slim) with SQL*Plus 21.3
Backend / ML	Python 3.14, Flask 3.0.3, psycopg2 2.9, scikit-learn 1.5+
Front end	React 19, Vite 8, tested in headless Chromium (Playwright)

Dataset. No real student data was used. `db/seed.py` generates a deterministic synthetic dataset: 180 students (30 in each of six departments), 18 companies, 25 drives, 517 applications and 177 offers, after which 95 students (52.8%) are Placed. The seed moves every application through legal status steps, so the audit trigger logged 1,177 status changes. Offers are made with a probability that rises with CGPA margin and skill count and falls with backlogs, so the dataset has a learnable signal. Every date is placed relative to the day the data is loaded (22 October 2026 for the evidence): completed drives fall in the past six months, open drives still take applications, and each application, interview round, offer and status change follows the one before it. The dates use their own random generator, so who is placed, and therefore the model, is identical however the dates are arranged.

Tests and measurements.

- **Smoke test** (`tests/smoke_test.py`): 48 checks through the API and directly against the database. It runs on a throwaway copy of the database, built and dropped by `tests/run_smoke_test.sh`.
- **API contract check** (`tests/api_contract_check.py`): parses `docs/API.md` and calls every documented endpoint, comparing status codes and response fields.
- **Browser checks** (`tests/single_link_check.js`, `tests/desktop_check.js`): headless Chromium signs in as a student and as the admin through the single address, in each way of running the system, and checks that no other origin is contacted. The Electron app is driven the same way.
- **Index benchmark** (`db/demos/index_benchmark.py`): a separate database of 50,000 students, 2,000 drives and 250,000 applications. Each query's median time over 7 warm runs is measured with and without the indexes.
- **SQL evidence**: every query, trigger and transaction was run in psql. Anything that changed data ran inside a transaction that was rolled back. The unedited output is in `docs/evidence/`.

2.9 Deployment

One address. The backend serves the built React app and the API from the same origin and port (http://localhost:8080) through waitress, a production WSGI server that also runs on Windows [17]. Any path that is not a file returns `index.html`, so refreshing works on every page, and the browser needs no CORS. The ML service listens only on the machine itself (or the private container network), and the backend calls it there.

One command. `docker compose up -d --build` [18] starts three containers: Postgres, the app and the ML service. Only the app publishes a port, and only on 127.0.0.1. On the first start the app container creates the tables, loads the synthetic data and generates a JWT secret into its own volume, and the ML container trains the model from the database. In this project's run, this took 235 seconds from nothing and gave the same model results as Section 3.6. `start.ps1` and `start.sh` do the same with local Python environments; they check that Docker is running and say so clearly if it is not.

Desktop app. A small Electron app [19] (Figure 4) checks Docker, offers Retry if it is not running, starts the same stack with a loading screen, shows PlaceIQ in its window and stops the stack when closed. It is packaged with electron-builder as a portable Windows .exe. It needs Docker Desktop and does not bundle it.

Sharing. `scripts/share.ps1` opens a Cloudflare quick tunnel [20] (free, no account) and prints a temporary https address. In this public mode the backend refuses to start without a strong secret, the documented demo passwords are replaced by private ones kept in a git-ignored file, and logins are rate-limited to 10 per 5 minutes per visitor. `DEPLOY_FREE.md` describes a permanent free option (Render with Neon Postgres) and its limits, as steps for the team to take; it was not deployed.

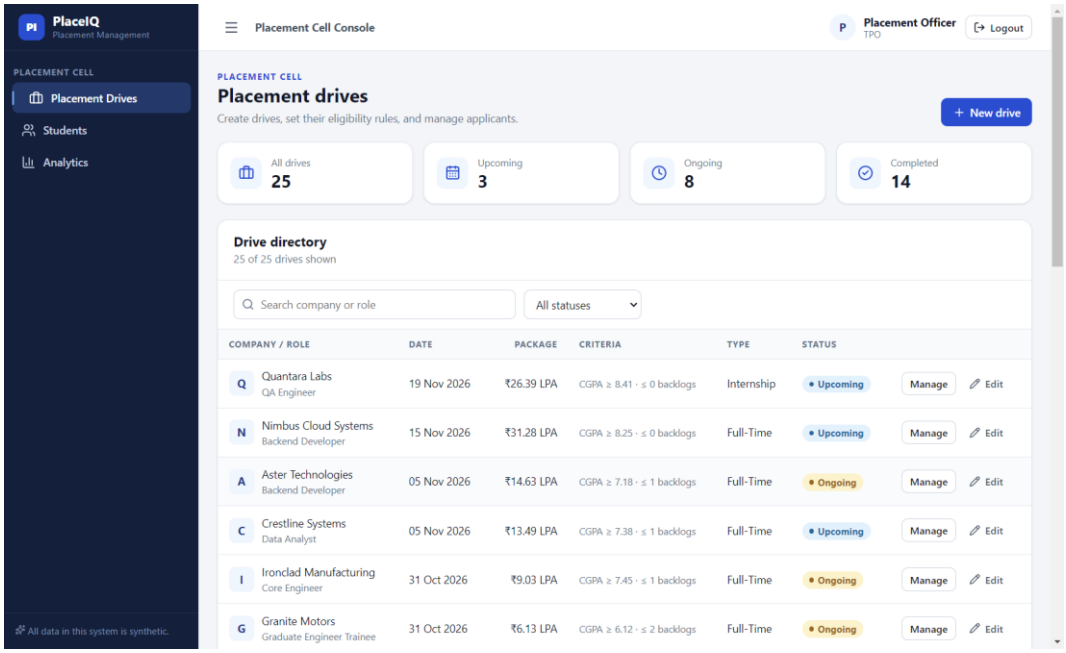


Figure 4: The desktop app's window, showing the placement-office console

3 Results and Discussion

All figures in this chapter are copied from the evidence files in `docs/evidence/` (listed in `docs/evidence/INDEX.md`). Each file holds the unedited output of the run it describes.

3.1 Functional Verification

Table 5: Verification results

Check	Result	Evidence
End-to-end smoke test (API and database)	48 / 48 passed	smoke-test.md
API documentation vs. running backend (36 endpoints)	161 / 161 passed	raw/api-contract-check.txt
Date-ordering rules over all seeded data	10 rules, 0 violations	date-integrity.md
Single address, Docker stack (fresh start)	12 / 12 passed	raw/single-link-docker.txt
Single address, local start script	12 / 12 passed	raw/single-link-venv.txt
Single address, public Cloudflare link	12 / 12 passed; demo password refused; 429 after 10 logins	raw/share-public-check.txt
Packaged desktop app	8 / 8 passed; nothing left after quit	raw/desktop-app-check.txt
Oracle schema in SQL*Plus	82 objects, all VALID; no compilation errors	sqlplus-transcript.txt
Oracle rule triggers	All 7 intended ORA-200xx errors raised, no others	sqlplus-transcript.txt

3.2 Rules Enforced by the Database

Each rule was tested by sending a violating statement straight to PostgreSQL, bypassing the backend (`triggers.md`). Table 6 lists what was run and what the database answered. Each refused write left no row behind. For example, after the PL006 attempt, application 1 still had 0 offers.

Table 6: Rule violations sent directly to the database

Statement (summary)	Database response	Code
Student 8 (CGPA below cutoff) applies to drive 1	Student 8 is not eligible for drive 1	PL001
A student applies to drive 2, which is Completed	Drive 2 is Completed and no longer accepts applications	PL002
Student 18, who already accepted an offer, accepts offer 70	Student 18 has already accepted another offer	PL003
Insert an offer for application 1 (Rejected)	Application 1 is Rejected; an offer can only be recorded for a Selected application	PL006
Move application 1 from Rejected to Applied	Application 1 cannot move from Rejected to Applied	PL007
Move application 2 from Offer_Made to Shortlisted	Application 2 cannot move from Offer_Made to Shortlisted	PL007
Student 69 applies to drive 1 after its deadline was moved to yesterday	Applications for drive 1 closed on 2026-10-21	PL008

The function `fn_check_eligibility` was checked against an independently computed reason for four real student–drive pairs (`queries.md`). It returned true for the pair meeting every rule. It returned false for a CGPA of 7.99 against a cutoff of 8.25, for 2 backlogs against a maximum of 1, and for a CSE student applying to a drive open only to AIML and ECE. Calling

`sp_close_drive` on drive 24 (Brightwave BPO, Full Stack Developer) rejected its 37 pending applicants (27 Applied, 5 Shortlisted, 5 Interview_Scheduled) and marked it Completed, and the audit trigger logged all 37 changes. The demonstration was then rolled back, and the drive was again Ongoing.

Date integrity. Ten ordering rules were counted over all seeded data (`date-integrity.md`). Applications are on or before their deadline and never in the future; rounds come after the application; finished rounds and offers are in the past; each offer follows the final round; and status changes come after the application and in the order they were logged. Every rule had 0 violations. The database also refused, with PL008, an application after a deadline and one dated an hour ahead, and the CHECK refused a deadline after the drive date.

3.3 Transactions and Concurrency

The offer transactions were run through the real backend with the database state read before and after each call (`transaction.md`):

- **Accept**: offer 178 went from Pending to Accepted with a joining date of 2026-12-21, and the history went from [Created] to [Created, Accepted] in one request.
- **Decline** of a student's only offer: the offer became Declined and `placement_status` returned from Placed to Not Placed, together with the history row.
- **Refused acceptance** (PL003): the backend answered HTTP 409, and the before and after states were identical. The transaction left nothing behind.
- **Two simultaneous acceptances** from two database connections: connection B was still blocked on the student-row lock 1.5 s later. When A committed, B was immediately refused with PL003. Offer 180 ended Accepted and offer 181 stayed Pending, exactly one acceptance.

At the SQL level (`transaction-sql.md`), the same three statements were run by hand and then rolled back, and all three changes disappeared together. In a second run a CHECK-constraint violation in the third statement aborted the transaction: PostgreSQL refused every following statement, the final COMMIT became a rollback, and the first two updates were undone. This is atomicity in practice [2], [10].

3.4 Placement Analytics

Table 7 is the output of `department_stats_view` on the seeded data. Electrical & Electronics has the highest placement rate (60.0%) and Computer Science the lowest (43.3%). Because the data is synthetic, these figures describe the generator's choices, not any real college.

Table 7: Placement by department (department_stats_view)

Department	Students	Placed	Placed %
Electrical & Electronics	30	18	60.00
Artificial Intelligence & ML	30	17	56.67
Information Science Engineering	30	16	53.33
Electronics & Communication	30	16	53.33
Mechanical Engineering	30	15	50.00
Computer Science Engineering	30	13	43.33

Table 8: Top recruiters by offers (company_wise_stats_view)

Company	Offers	Average package (LPA)
Coral Bay Solutions	80	6.89
Everstone Infra	41	7.14
Solstice Retail Tech	24	5.51
Bluepeak Software	8	14.62
Meridian Consulting	8	10.14

Offers per month (from `placement_timeline_view`) were 114, 36, 1, 7 and 19 for May to September 2026, a running total of 177 (Figure 6). An earlier version of the data exposed a defect in this view. It only had rows for months with offers, so with no offers in January 2026 `LAG()` compared February with December and reported a fall of 98. The view now generates every month between the first and last offer (`generate_series` in PostgreSQL, `ADD_MONTHS` with `CONNECT BY LEVEL` in Oracle) and left-joins the counts, so January appears as 0 and February's change is +16. The fix was re-verified on both databases. It shows a general pitfall of `LAG()`: it reads the previous row, not the previous calendar period.

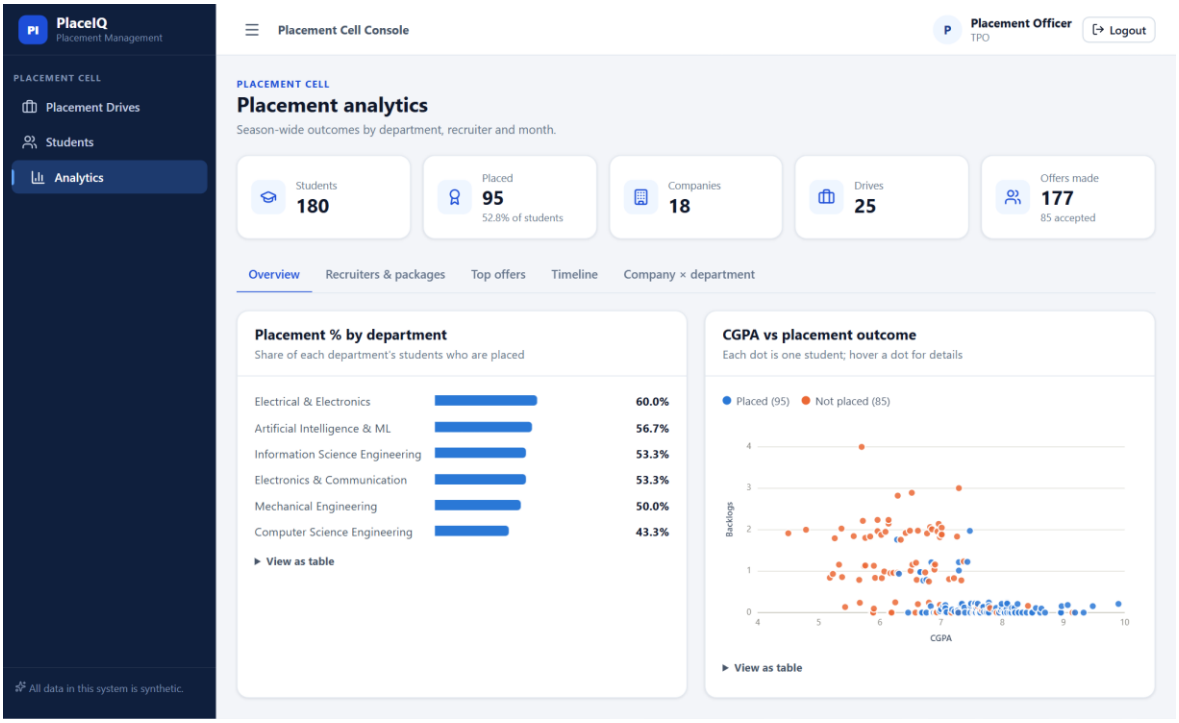


Figure 5: Analytics: placement % by department and CGPA against placement outcome

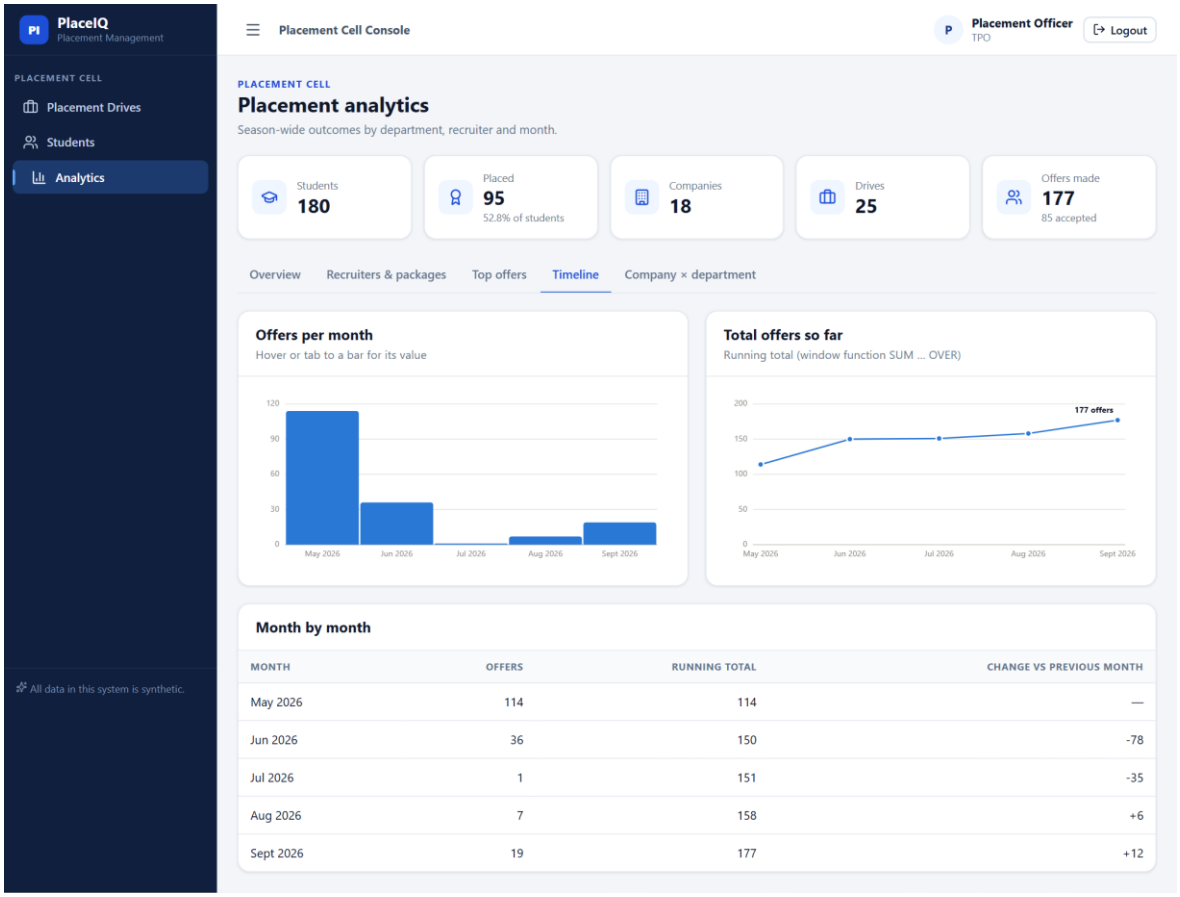


Figure 6: Offers per month and running total, from the window-function view

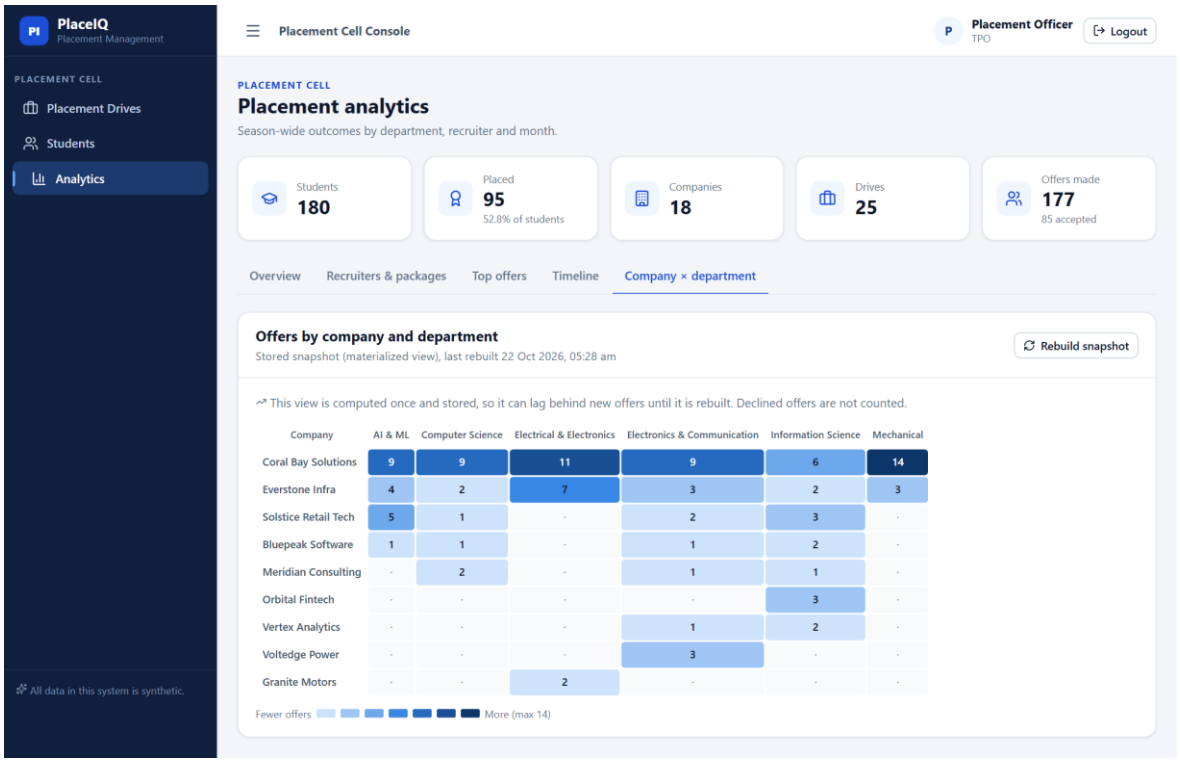


Figure 7: Offers by company and department, read from the materialized view

3.5 Effect of the Indexes

Table 9 gives the median execution time of five of the application's lookups on the 250,000-application benchmark database, with and without the indexes (`index-benchmark.md`, which also contains every query plan).

Table 9: Index benchmark (median of 7 warm runs)

Query	Index used	With	Without	Speed-up
Applicants for one drive	idx_application_drive	0.13 ms	10.31 ms	81.2×
CSE students with CGPA ≥ 9.7	idx_student_cgpa + idx_student_dept	0.62 ms	6.15 ms	9.9×
Count of students in one department	idx_student_dept	2.08 ms	7.20 ms	3.5×
Count of applications in one status	idx_application_status	4.56 ms	13.23 ms	2.9×
Drives of one company	idx_drive_company	0.07 ms	0.12 ms	1.8×

The gain depends on selectivity. Listing one drive's applicants touches 125 of 250,000 rows. Through the index PostgreSQL read 127 buffer pages instead of scanning all 1,966, an 81× speed-up. For the CGPA and department query the planner combined two indexes with a `BitmapAnd` before reading the table. A filter that matches a sixth of the students (one department) or 3% of the applications (one status) still has to visit most pages, so the gain is smaller. The `drive.company_id` index barely helps (0.07 vs 0.12 ms) because the whole `drive` table is 27 pages. It is kept for the drive-to-company join, but the measurement does not justify it on speed alone. On the real 180-student dataset none of the indexes matters, which is why the benchmark uses a separate, larger database.

3.6 Placement-Probability Model

Table 10: Model results on the held-out test set

Measure	Value
Training / test rows	144 / 36 (test: 19 placed, 17 not placed)
Training accuracy	0.875
Test accuracy	0.750 (27 of 36 correct)
Test ROC-AUC	0.774
Precision / recall for 'placed'	0.750 (15 of 20) / 0.789 (15 of 19)
Confusion matrix (actual rows × predicted)	Not placed: 12 correct, 5 wrong; Placed: 15 correct, 4 wrong

The coefficients point the expected way. CGPA has the largest positive weight (+1.496 log-odds per standard deviation), backlogs the largest negative (−1.136), and the skill count is positive (+0.819). Department effects are smaller. Figure 8 shows the estimate as a student sees it.

These results must be read with care. The test set has only 36 students, so one prediction changes accuracy by about 2.8 percentage points, and another random split could easily give 0.70 or 0.80. The data is synthetic, and the generator itself makes CGPA, backlogs and skills predictive, so the model largely recovers rules that were built in. The result shows that the pipeline works end to end (database, features, model, API, interface). It does not show that the model would predict real placements. For comparison, Kumar et al. [6] report Random Forest

as the most effective of the methods they compared on real student records. Choosing a model on real data would need a comparison of that kind.

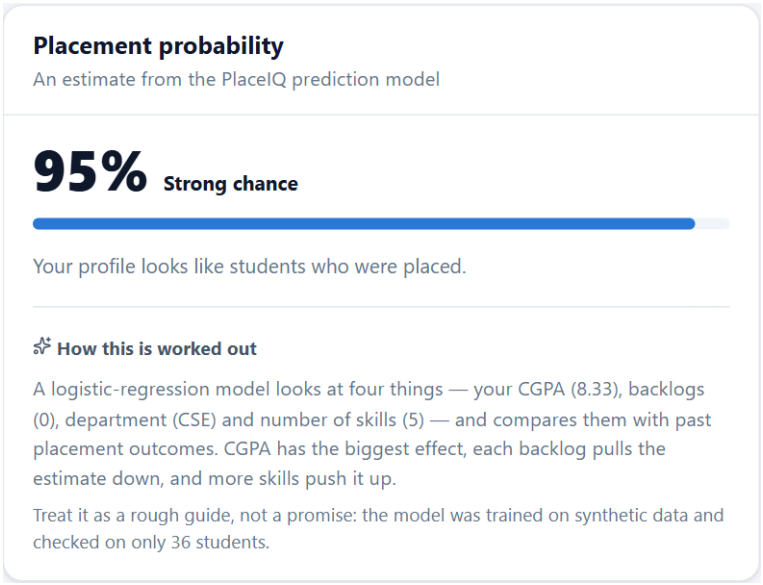


Figure 8: The placement-probability card on the student dashboard

4 Conclusion and Future Work

PlaceIQ shows that a placement office's rules can live in the database rather than in application code. Eligibility, the one-accepted-offer rule, offers only for selected applicants and forward-only status changes are all enforced by triggers, and each was shown to refuse a violating write sent directly to PostgreSQL. The offer transactions are atomic and remain correct under concurrent requests because the trigger locks the student's row. Window functions and a materialized view supply the analytics. The indexes were measured rather than assumed: up to 81× faster for selective lookups, and negligible on a tiny table. The same design runs on Oracle 21c with all 82 objects valid. The system runs from one address and starts with one command, as a desktop app or on a shareable link, all with free tools.

Limitations

- **Synthetic data.** All students, companies and outcomes are generated. The analytics describe the generator, not a real college.
- **Self-reported CGPA.** Students enter their own CGPA and backlogs at sign-up, and eligibility depends on them. A real deployment would need these values verified against the institution's records through an official integration, not entered by the student.
- **Small ML evaluation.** The model was tested on 36 synthetic students with a single split. Its accuracy is a rough figure, and its strength largely reflects how the data was generated.
- **Oracle port covers the database layer only.** The backend runs only against PostgreSQL. Mapping Oracle's −20001 to −20008 errors in the backend and the Oracle date arithmetic in the offer service were not built. On Oracle, the concurrent-acceptance behaviour rests on documented locking and was not demonstrated, because the demo is a single SQL*Plus session. Oracle has no `CONCURRENTLY` refresh option.
- **Deployment needs Docker Desktop.** The desktop app and the one-command start need Docker Desktop installed; the app does not bundle it, and the .exe is unsigned, so Windows SmartScreen may warn.
- **Temporary public link.** The Cloudflare quick-tunnel address lasts only while the laptop runs it. The permanent free hosting in `DEPLOY_FREE.md` was not deployed, and free tiers have cold starts.
- **Placement policy.** A placed student may keep applying to other drives. Many colleges restrict this, and the rule would be one more trigger.

Future Work

- Verify academic records through an official institutional data feed instead of self-reported values.
- Evaluate the model on real, consented and anonymized placement records with cross-validation, and compare it with other classifiers such as Random Forest [6].
- Add recruiter accounts, e-mail notifications and document uploads (resumes, offer letters).
- Make the placement policy configurable (for example, how many offers a placed student may hold) and enforce it with a trigger.

- Complete an Oracle-backed deployment of the backend and demonstrate concurrent acceptance there.
- Schedule the materialized-view refresh, or refresh it when offers change, instead of on demand only.

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Appendix A Code Listings

The listings below are read from the source files when this report is generated, with their real line numbers, so they match the submitted code exactly.

A.1 Rule PL003: one accepted offer per student, with the row lock

`db/schema.sql`, lines 552–585.

```
552  CREATE OR REPLACE FUNCTION trg_fn_offer_single_acceptance()
553  RETURNS TRIGGER AS $$
554  DECLARE
555      v_student_id student.student_id%TYPE;
556  BEGIN
557      IF NEW.status <> 'Accepted' THEN
558          RETURN NEW;
559      END IF;
560
561      SELECT student_id INTO v_student_id
562        FROM application WHERE application_id = NEW.application_id;
563
564      PERFORM 1 FROM student WHERE student_id = v_student_id FOR UPDATE;
565
566      IF EXISTS (
567          SELECT 1
568            FROM offer o
569            JOIN application a ON a.application_id = o.application_id
570           WHERE a.student_id = v_student_id
571             AND o.status = 'Accepted'
572             AND o.offer_id IS DISTINCT FROM NEW.offer_id
573      ) THEN
574          RAISE EXCEPTION 'Student % has already accepted another offer', v_student_id
575              USING ERRCODE = 'PL003';
576      END IF;
577
578      RETURN NEW;
579  END;
580  $$ LANGUAGE plpgsql;
581
582  CREATE TRIGGER trg_offer_before_accept
583  BEFORE INSERT OR UPDATE OF status ON offer
584  FOR EACH ROW
585  EXECUTE FUNCTION trg_fn_offer_single_acceptance();
```

A.2 Rule PL007: the status-transition state machine

`db/schema.sql`, lines 634–661.

```
634  CREATE OR REPLACE FUNCTION trg_fn_application_status_transition()
635  RETURNS TRIGGER AS $$
636  DECLARE
637      v_allowed BOOLEAN;
638  BEGIN
639      v_allowed := CASE OLD.status
640          WHEN 'Applied'             THEN NEW.status IN ('Shortlisted', 'Interview_Scheduled', 'Rejected')
641          WHEN 'Shortlisted'         THEN NEW.status IN ('Interview_Scheduled', 'Selected', 'Rejected')
642          WHEN 'Interview_Scheduled' THEN NEW.status IN ('Selected', 'Rejected')
643          WHEN 'Selected'            THEN NEW.status IN ('Offer_Made', 'Rejected')
644          ELSE FALSE   -- Offer_Made and Rejected are final
645      END;
646
647      IF NOT v_allowed THEN
648          RAISE EXCEPTION 'Application % cannot move from % to %',
649              NEW.application_id, OLD.status, NEW.status
650              USING ERRCODE = 'PL007';
651      END IF;
652
653      RETURN NEW;
654  END;
655  $$ LANGUAGE plpgsql;
656
657  CREATE TRIGGER trg_application_status_transition
658  BEFORE UPDATE OF status ON application
659  FOR EACH ROW
660  WHEN (OLD.status IS DISTINCT FROM NEW.status)
661  EXECUTE FUNCTION trg_fn_application_status_transition();
```

A.3 Stored procedure sp_close_drive

`db/schema.sql`, lines 721–751.

```
721  CREATE OR REPLACE PROCEDURE sp_close_drive(
722      p_drive_id        INTEGER,
723      INOUT p_rejected_count INTEGER DEFAULT NULL
724  )
725  LANGUAGE plpgsql
726  AS $$
727  DECLARE
728      v_status drive.status%TYPE;
729  BEGIN
730      -- Lock the drive row so two admins closing the same drive at once can't
731      -- both pass the "already closed?" check below.
732      SELECT status INTO v_status FROM drive WHERE drive_id = p_drive_id FOR UPDATE;
733
734      IF NOT FOUND THEN
735          RAISE EXCEPTION 'Drive % does not exist', p_drive_id USING ERRCODE = 'PL004';
736      END IF;
737
738      IF v_status IN ('Completed', 'Cancelled') THEN
739          RAISE EXCEPTION 'Drive % is already %', p_drive_id, v_status USING ERRCODE = 'PL005';
740      END IF;
741
742      UPDATE application
743         SET status = 'Rejected'
744       WHERE drive_id = p_drive_id
745         AND status IN ('Applied', 'Shortlisted', 'Interview_Scheduled');
746
747      GET DIAGNOSTICS p_rejected_count = ROW_COUNT;
748
749      UPDATE drive SET status = 'Completed' WHERE drive_id = p_drive_id;
750  END;
751  $$;
```

A.4 Window-function view with the month series (placement_timeline_view)

`db/schema.sql`, lines 795–812.

```
795  CREATE VIEW placement_timeline_view AS
796  WITH monthly AS (
797      SELECT date_trunc('month', offer_date)::DATE AS month, COUNT(*) AS offers
798      FROM offer
799      GROUP BY date_trunc('month', offer_date)
800  ),
801  all_months AS (
802      SELECT gs::DATE AS month
803      FROM (SELECT MIN(month) AS first_month, MAX(month) AS last_month FROM monthly) b,
804           generate_series(b.first_month, b.last_month, INTERVAL '1 month') gs
805  )
806  SELECT
807      m.month,
808      COALESCE(x.offers, 0) AS offers_in_month,
809      SUM(COALESCE(x.offers, 0)) OVER (ORDER BY m.month) AS cumulative_offers,
810      COALESCE(x.offers, 0) - LAG(COALESCE(x.offers, 0)) OVER (ORDER BY m.month) AS change_from_prev_month
811  FROM all_months m
812  LEFT JOIN monthly x ON x.month = m.month;
```

A.5 The offer-acceptance transaction (backend)

`backend/services/offer_service.py`, lines 65–115.

```
65  def accept_offer(offer_id: int, requesting_user: dict) -> dict:
66      with db.get_connection() as conn:
67          cur = conn.cursor(cursor_factory=psycopg2.extras.RealDictCursor)
68          try:
69              offer_row = _load_offer_owner(cur, offer_id)
70              if offer_row is None:
71                  raise OfferNotFound(f"Offer {offer_id} does not exist")
72              _authorize(requesting_user, offer_row["student_id"])
73              student_id = offer_row["student_id"]
74
75              # --- Statement 1: flip the Offer itself -----------------------
76              # WHERE status='Pending' + RETURNING doubles as an atomic guard:
77              # if two requests race to accept/decline the same offer, only
78              # the first UPDATE finds a matching row — the second gets back
79              # no row and this function raises instead of silently
80              # double-processing the offer.
81              cur.execute(
82                  """
83                  UPDATE offer
84                  SET status = 'Accepted',
85                      joining_date = COALESCE(joining_date, offer_date + INTERVAL '60 days')
86                  WHERE offer_id = %s AND status = 'Pending'
87                  RETURNING offer_id
88                  """,
89                  (offer_id,),
90              )
91              if cur.fetchone() is None:
92                  raise OfferNotActionable(
93                      f"Offer {offer_id} is not Pending (already actioned or invalid)"
94                  )
95
96              # --- Statement 2: update Student status ------------------------
97              cur.execute(
98                  "UPDATE student SET placement_status = 'Placed' WHERE student_id = %s",
99                  (student_id,),
100             )
101
102             # --- Statement 3: append to the audit trail ---------------------
103             cur.execute(
104                 "INSERT INTO offer_history (offer_id, action, notes) VALUES (%s, 'Accepted', %s)",
105                 (offer_id, "Offer accepted by student"),
106             )
107
108             conn.commit()
109         except Exception:
110             conn.rollback()
111             raise
112         finally:
113             cur.close()
114
115     return {"offer_id": offer_id, "status": "Accepted"}
```

Appendix B Screenshots

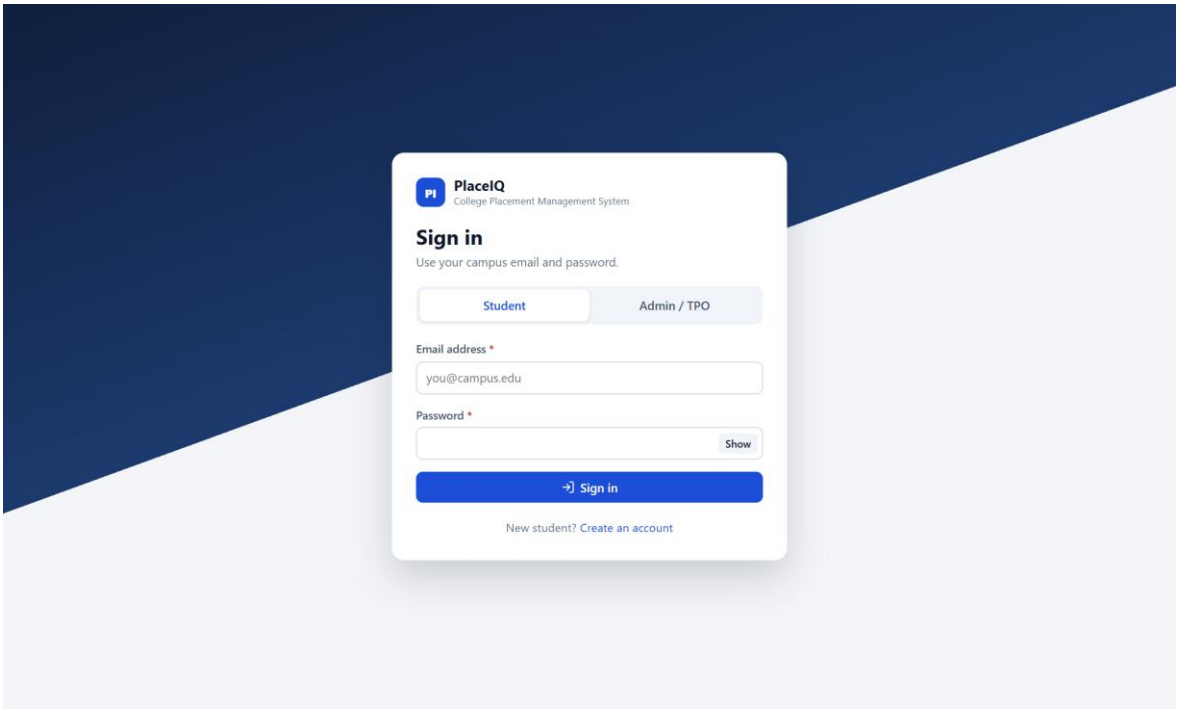


Figure 9: Sign-in page (student and placement-office roles)

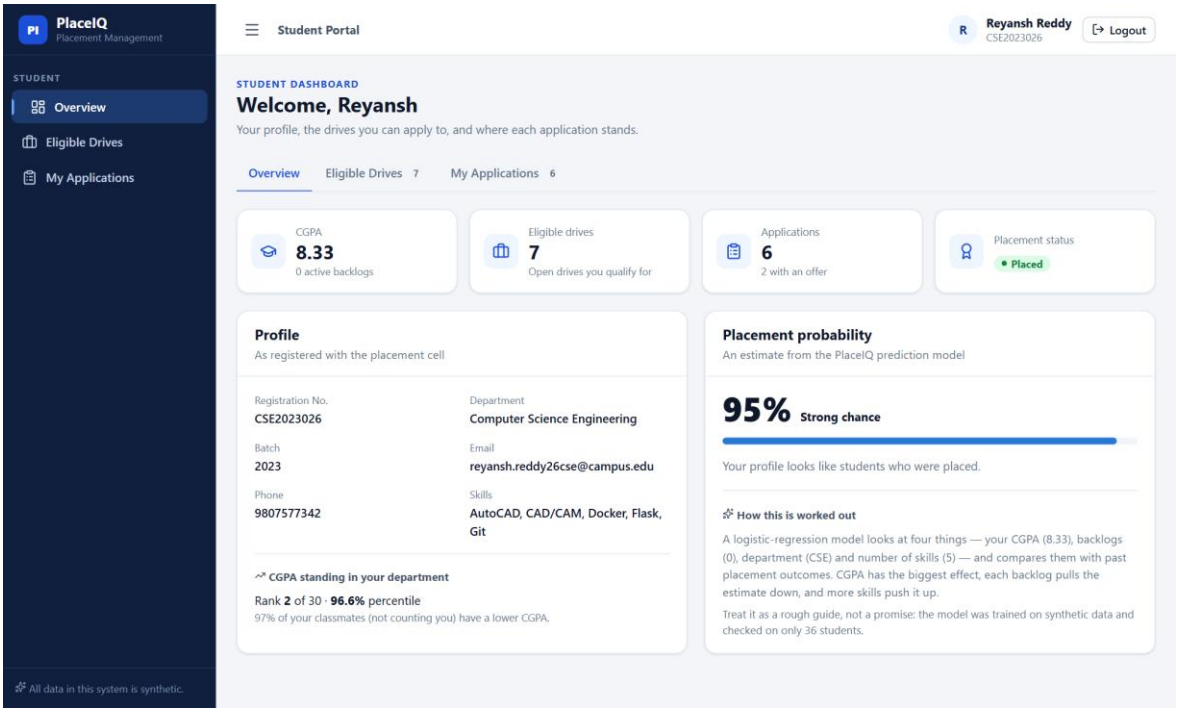


Figure 10: Student dashboard: profile, CGPA standing in the department, placement probability

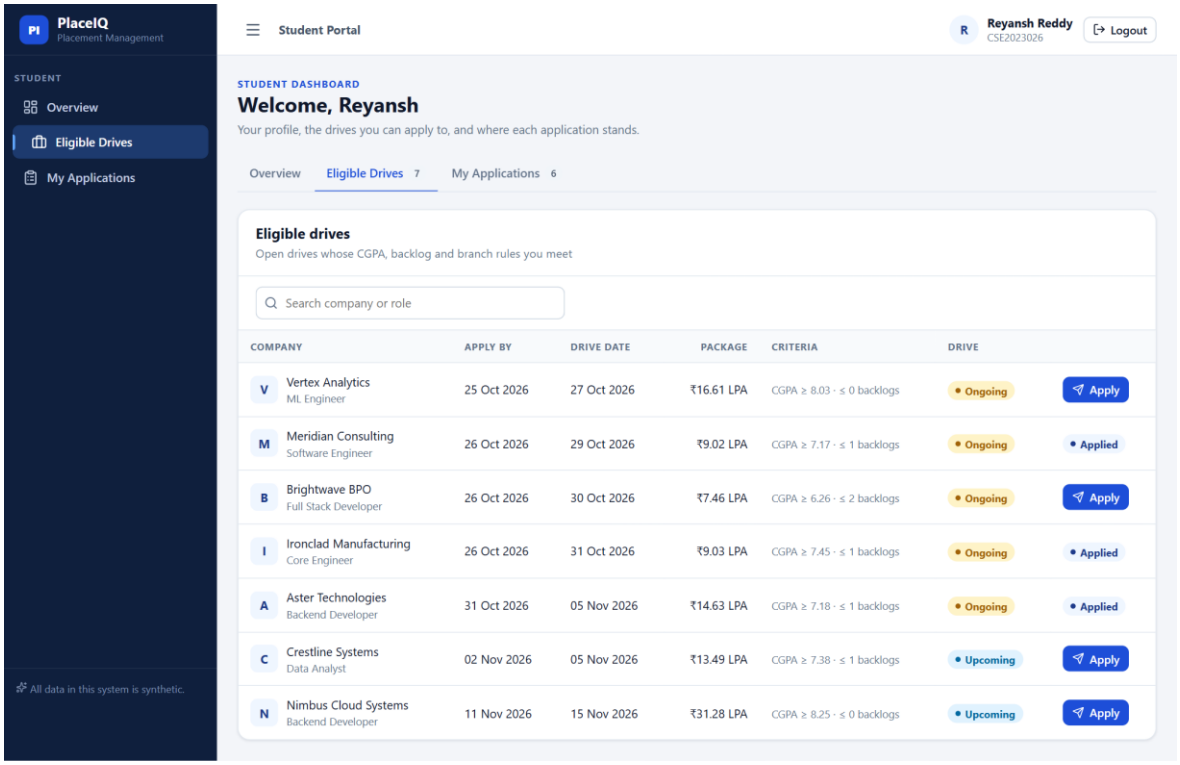


Figure 11: Eligible drives, from fn_get_eligible_drives

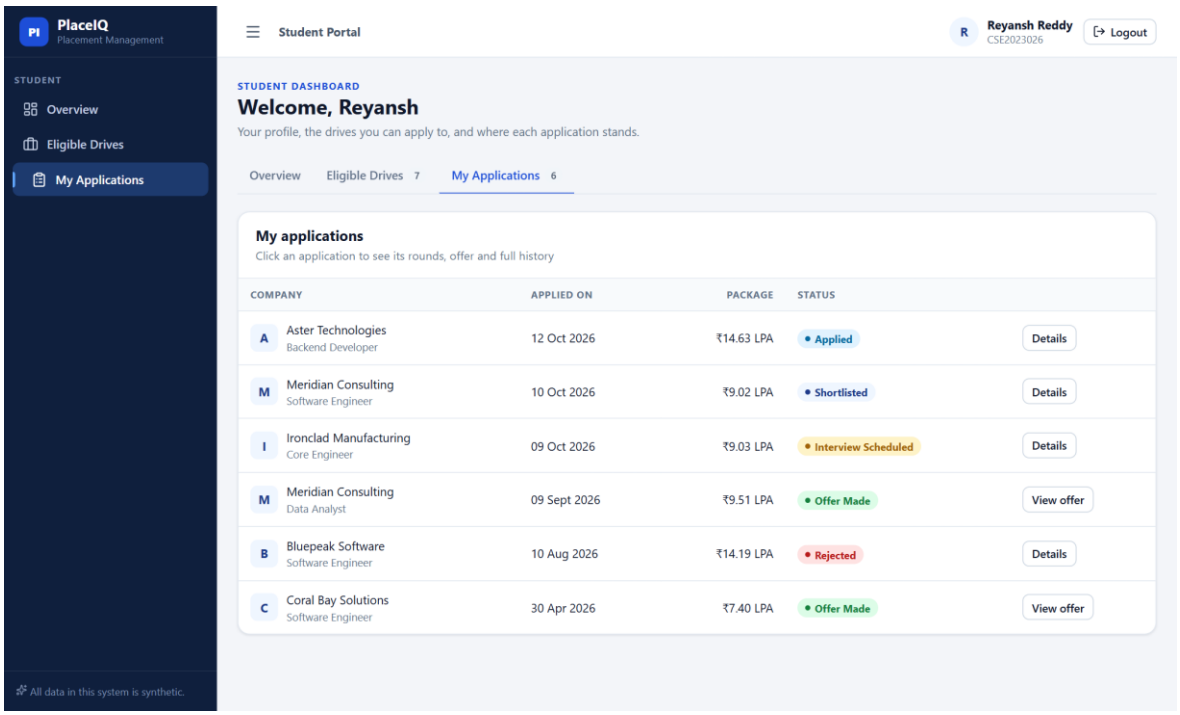


Figure 12: A student's applications at five different stages

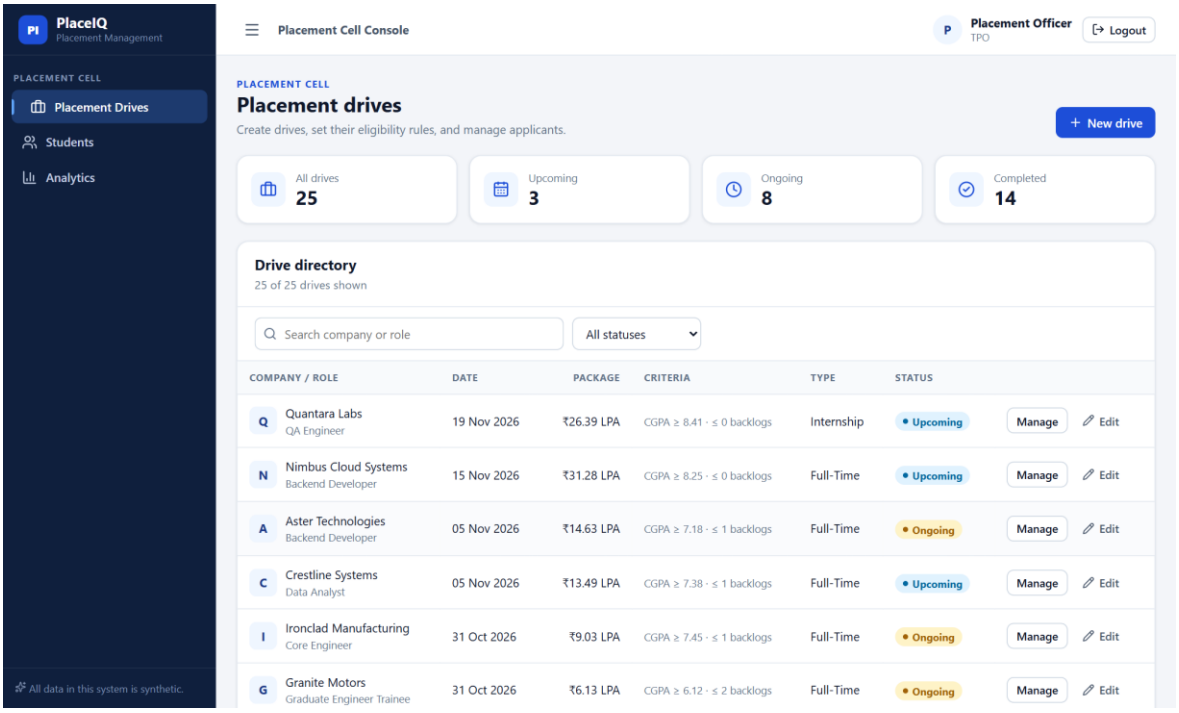


Figure 13: Placement office: list of drives

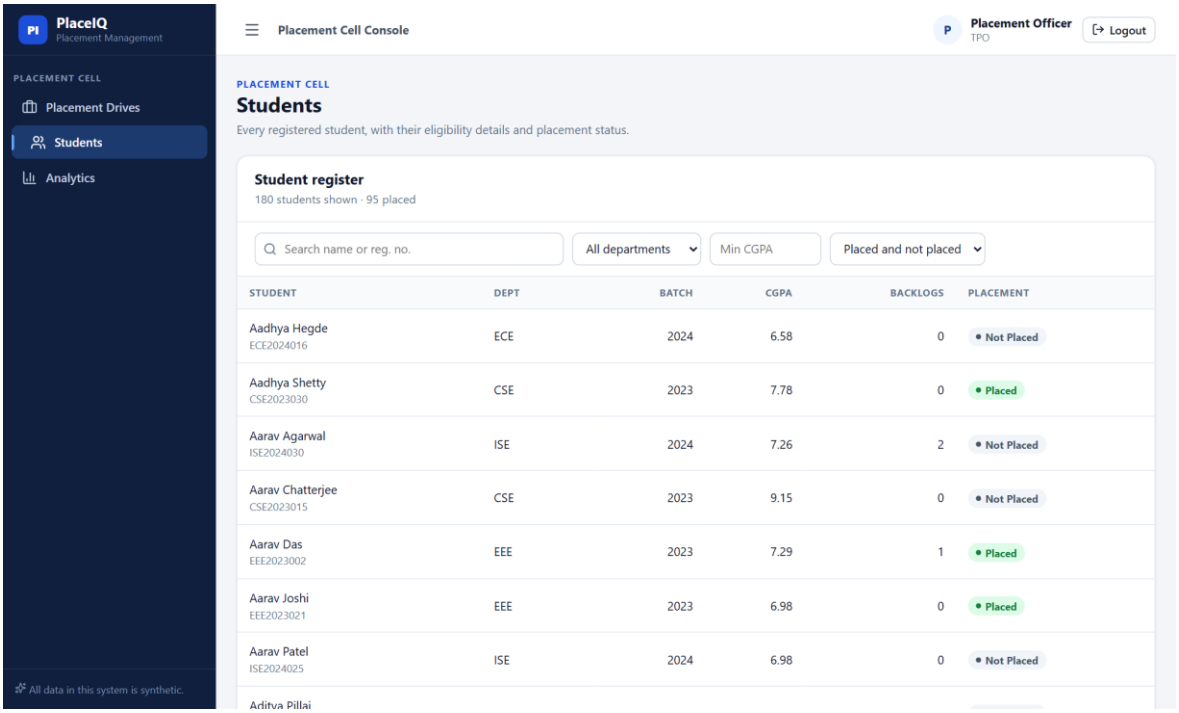


Figure 14: Placement office: student list

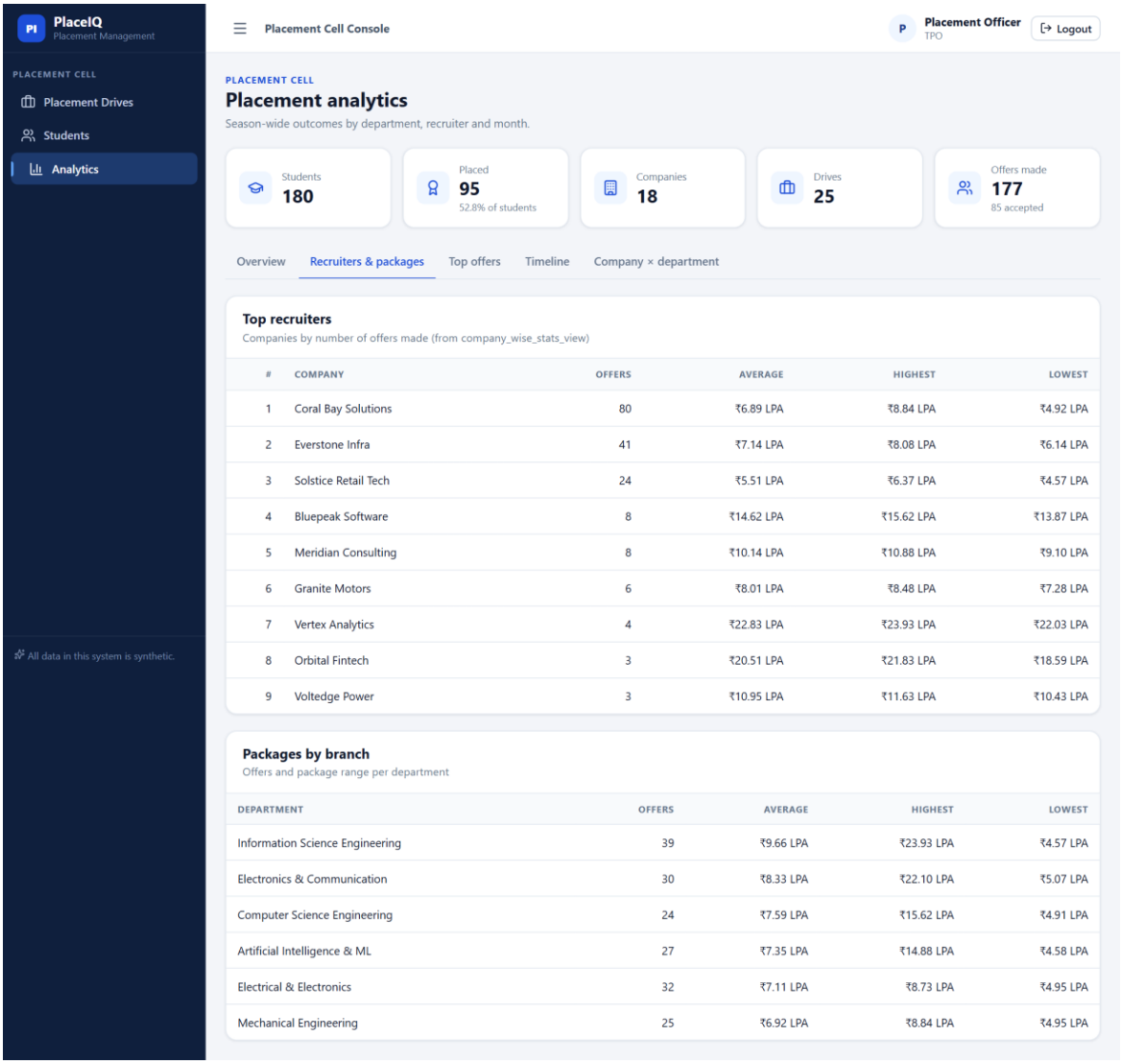


Figure 15: Analytics: recruiters and packages

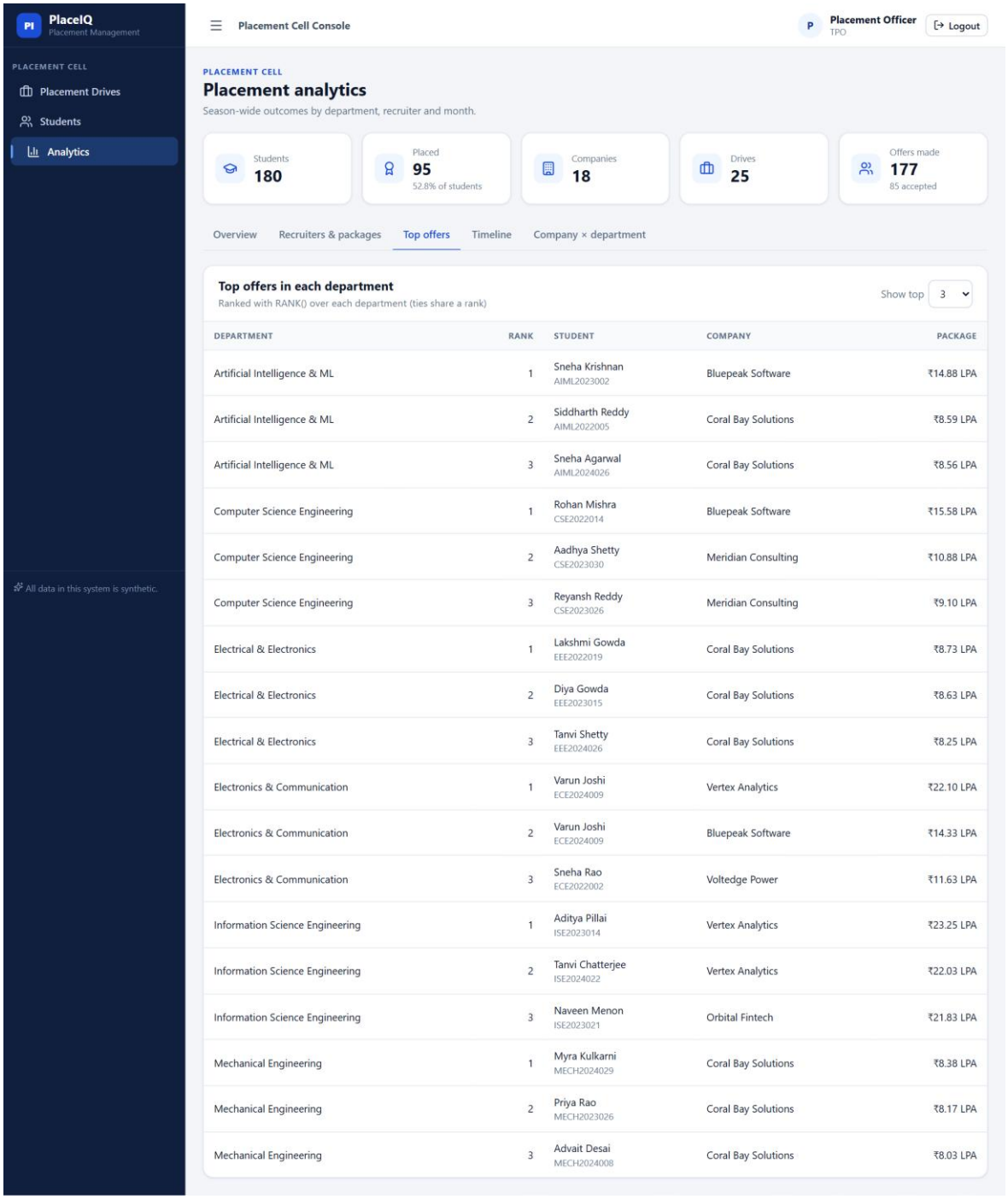


Figure 16: Analytics: top offers in each department (RANK)

Appendix C Evidence Files

The `docs/evidence/` folder of the repository holds the unedited output behind every result in this report. Its index (`docs/evidence/INDEX.md`) lists each file, the report section it supports and the script that regenerates it. Among them are the SQL*Plus transcript of the Oracle run, the full EXPLAIN ANALYZE plans of the benchmark, and the schema summary read from the PostgreSQL catalog.